Final Project Code Auditing and Improvement

Weighting	70% of final module grade
Submission format	Individual or small groups (declare on submission)
Language	Python (scikit-learn, PyTorch, or TensorFlow)
Deliverables	Audit report (PDF) + improved code (.py or .ipynb) + results summary

1. Overview

You are provided with a deliberately flawed Python machine learning pipeline. Your task is to act as a critical peer reviewer: identify what is wrong, explain why it matters, improve the pipeline, and demonstrate that your changes lead to more reliable, trustworthy, and methodologically sound results.

This assignment assesses your ability to read, interpret, and critically evaluate existing ML code: a skill that is as important in practice as writing new code from scratch. You are expected to engage with methodological decisions (e.g., data splitting, evaluation design, metric selection), not just syntax errors.

This is a capstone-style assignment. You are expected to demonstrate both conceptual understanding and the ability to work with existing machine learning code. The emphasis is on critical analysis and reasoning, not on building a system from scratch.

The audit report (Part A) is the most important component of this assignment. A strong analytical report that clearly identifies and explains key issues will be graded more highly than a fully working implementation with a weak or superficial analysis.

2. Problem type selection

For this assignment, you will select one of the following problem types:

- supervised classification
- supervised regression
- unsupervised learning

Based on your selection, you will be assigned a corresponding machine learning pipeline for auditing. The provided pipeline will reflect the chosen problem type, and your analysis, evaluation strategy, and improvements must be appropriate to that setting.

Think of this assignment as reviewing and correcting a flawed machine learning system, rather than building one from scratch. The goal is not to replace the pipeline entirely, but to identify and correct its most critical weaknesses.

3. Assignment setup

Before starting, you must register your project group on ISIS. During registration, indicate:

- Whether you are working individually or in small groups
- Whether you are doing supervised classification, regression, or unsupervised learning
- Whether you are taking the assignment graded (Benotet) or pass/fail (Unbenotet, only for MSc Human Factors students)

After registration, you will be assigned a corresponding codebase for auditing.

4. The Provided Code

The code provided for this assignment is based on anonymized student submissions from previous courses taught by the instructor and has been modified where necessary.

The pipelines are functional and may produce plausible results, but they contain real-world methodological flaws and design issues commonly encountered in practice. These may include:

- Data leakage
- Improper train/validation/test use
- Inappropriate metric selection
- Poor preprocessing design
- Ignored class imbalance, where relevant
- Overfitting or weak validation strategy
- Missing reproducibility controls
- Weak code readability (unclear variables, magic numbers, or redundant operations)

You are not expected to find every issue, but you should identify and analyze the most impactful ones.

5. Important Clarifications

- The provided code does not represent a correct or recommended solution.
- The assignment does not evaluate the work of previous students.
- The code is used solely as a learning artifact.
- Pipelines may differ in structure and difficulty, but all contain significant issues suitable for analysis.
- The goal is not to rewrite everything, but to critically analyze and improve the existing system.

Many real-world ML systems produce convincing results despite being methodologically flawed. This assignment trains you to recognize and correct such cases.

6. Tasks

Part A: Audit Report (18 marks | max. 1,500 words excluding tables/figures)

Your report should include:

1. Task and dataset understanding

Briefly describe:

- The type of ML task
- The dataset structure and characteristics
- Potential risks (e.g., imbalance, missing data, bias, leakage risks)

2. Issue identification

Identify the most important flaws in the provided pipeline. For each issue, explain:

- What the problem is
- Where it appears in the code
- Why it matters
- Its likely impact on performance, validity, reliability, or interpretation

Where possible, quantify the impact, for example by showing how leakage inflates performance.

3. Severity classification

Classify each issue as:

- Critical: invalidates results or would cause serious harm in deployment
- Major: reduces reliability, validity, performance, or interpretability
- Minor: affects readability, maintainability, or clarity

Justify each classification.

4. Methodological critique

Discuss the overall pipeline design:

- Is the evaluation design correct?
- Are the metrics appropriate?
- Is preprocessing handled correctly?
- Is the modelling approach suitable?
- Are the conclusions supported by the results?

5. Improvement plan

Before presenting your improved code, describe the fixes you intend to make and justify them. This should read as a design plan, not a code walkthrough.

Part B: Improved Implementation (18 marks)

Produce an improved pipeline that:

- Fixes all Critical and Major issues
- Uses correct evaluation (split or cross-validation)
- Avoids data leakage
- Applies preprocessing correctly
- Uses appropriate evaluation metrics
- Handles class imbalance where relevant
- Ensures reproducibility (random seeds)
- Includes clear comments explaining decisions
- Runs end-to-end without modification

The goal is a pipeline that is methodologically sound and professionally acceptable, not unnecessarily complex.

Part C: Results Summary and Reflection (9 marks | max. 500 words)

Append a short results section to your audit report.

Include:

- Comparison table: Compare original vs improved pipeline under identical evaluation conditions.
- Interpretation: Explain:
    - What changed
    - Why it changed
    - Which fixes mattered
- Reflection: Briefly reflect on what you learned.
    - What surprised you?
    - What would you do differently if starting again?
    - How did the assignment change your understanding of ML evaluation?

7. Submission Requirements
Submit one .zip file containing:
- audit_report.pdf
- improved_pipeline.py or improved_pipeline.ipynb
- requirements.txt
- README.txt

Do not submit the original provided code, model weights, or large data files.

The README should include:
- group member information (names, matriculation numbers, study program)
- brief instructions for running the code

Group Attribution
For group submissions, each section of the assignment must clearly indicate which group member(s) contributed to that part. This should be placed at the beginning of each section using a simple format, for example: Contributors: Name A, Name B

All group members are expected to be familiar with the entire submission, but this attribution ensures transparency in contribution.

Reporting Requirements
All written components of this assignment must be presented in full sentences and structured paragraphs. Bullet points may be used sparingly (e.g., for tables or summaries), but they do not replace proper explanations. Listing issues without explanation is not sufficient.

At a master's level, you are expected to:
- explain your reasoning clearly and precisely
- justify methodological decisions
- connect evidence to conclusions
- present arguments in a structured and coherent way

Submissions consisting primarily of bullet points or fragmented statements will not meet the expected standard.

Do not submit the original provided code, model weights, or large data files.

8. Academic Integrity
AI coding assistants, including ChatGPT, GitHub Copilot, or similar tools, may not be used to generate, debug, rewrite, or improve the submitted code or report.

You may use:
- Course materials
- Textbooks
- Official library documentation
- General conceptual discussions with peers

All submitted work must be your own. If working in a group, all members are responsible for understanding the full submission.

The provided code has been anonymized. Any attempt to identify previous students or locate original submissions is inappropriate.

9. Guidance and Tips

- Run and carefully analyze the provided code before writing anything.
- Focus on impactful issues, not minor quirks.
- The audit matters more than the code.
- Ensure fair comparisons (same data, same evaluation conditions).
- Lower accuracy is acceptable if results are more reliable and methodologically sound. You must explain why.